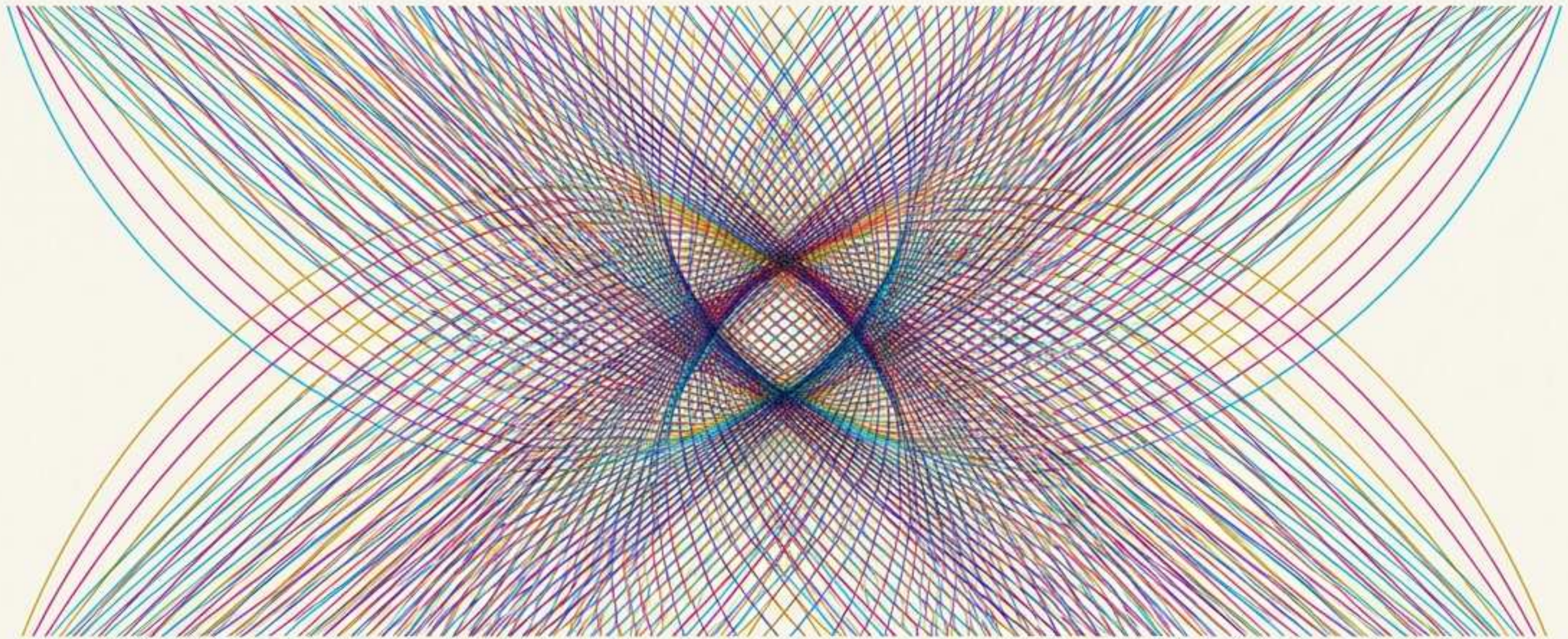


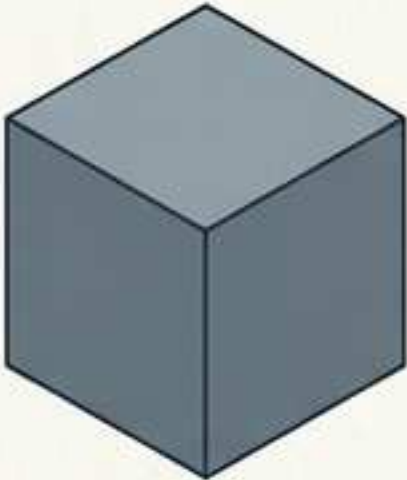
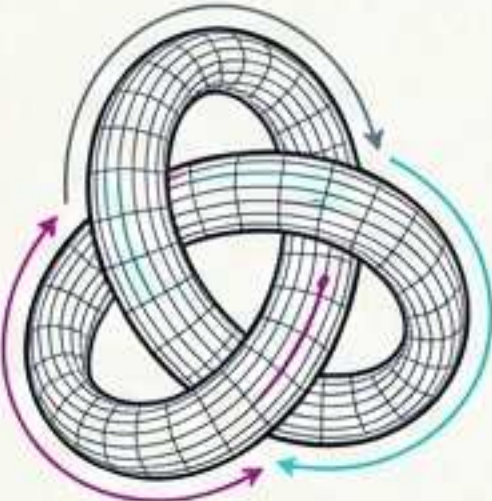
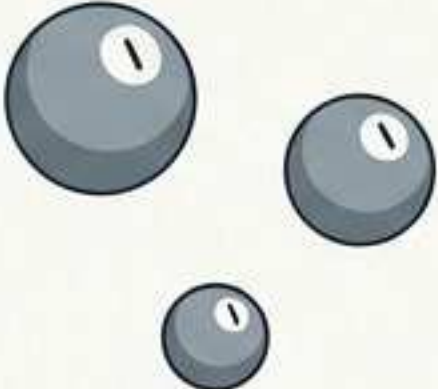
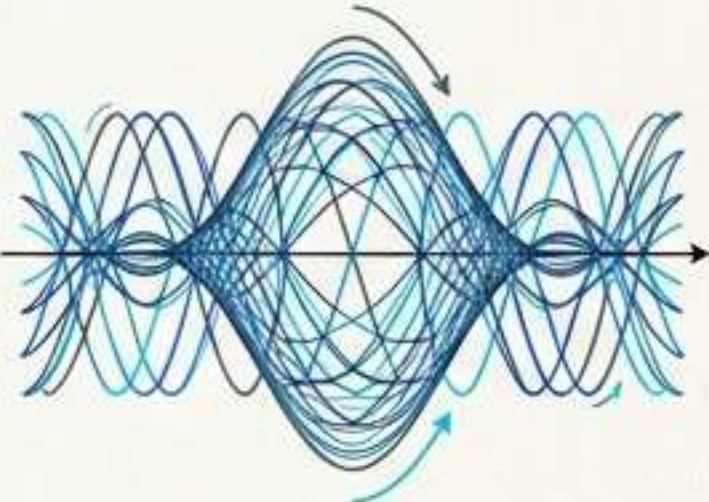
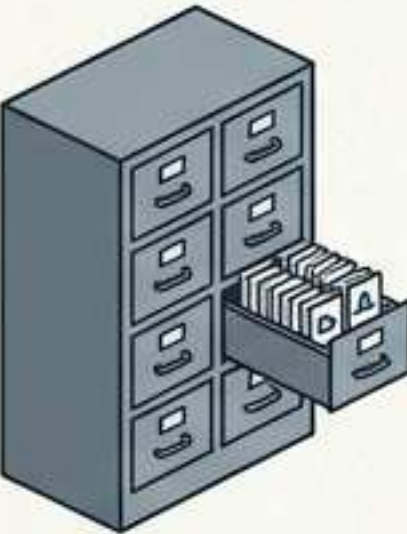
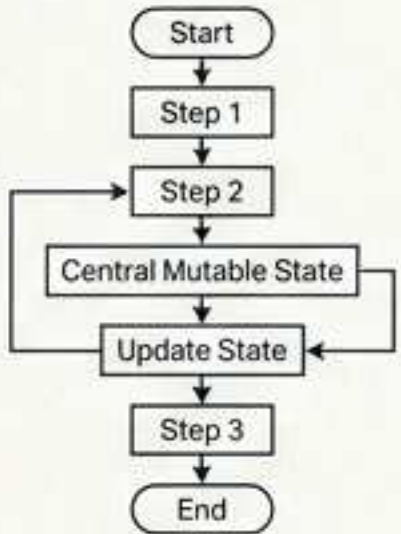
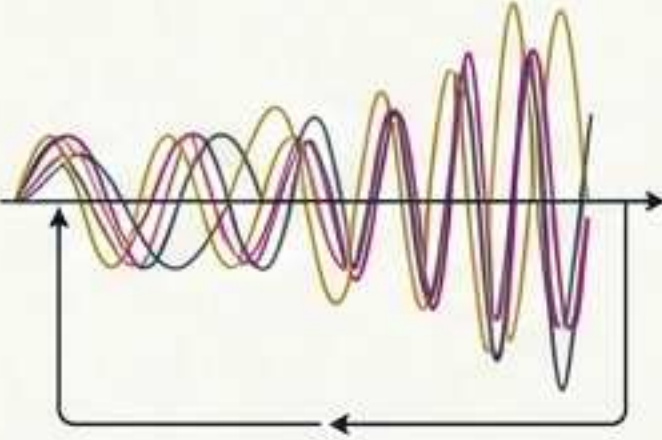
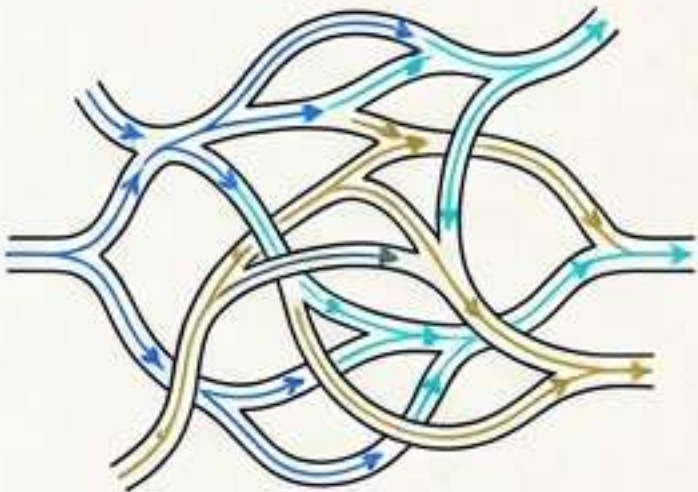


Process, Projection, and Propagative Reality

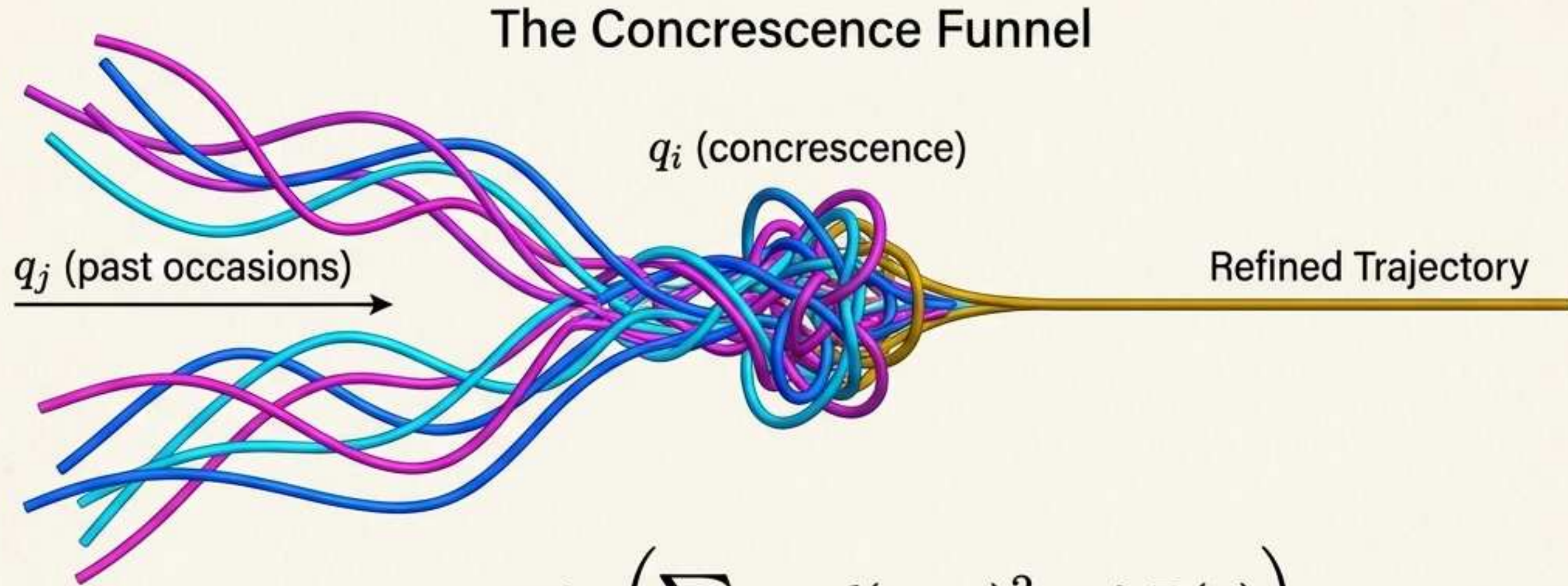
A unified structural grammar for recursive stabilization across physics, cognition, and computation.

Things are not things. They are recursively stabilized trajectories.

The Substance Fallacy vs. Process Reality

	Substance Ontology		Process Ontology	
Identity				
Matter	Primitive, static sameness over time.	Stable recoverability under constrained transformation.	Miniature billiard balls in empty space.	Localized attractors of field coherence (standing waves)
Memory				
Computation	Inert symbols stored in a spatial container.	Sequential execution of central mutable state.	Reconstructed resonance and recursive phase alignment.	Constrained propagation across admissibility manifolds.

Pillar 1: Whiteheadian Occasions & The Geometry of Inheritance



$$q_i = \arg \min_{q \in Q} \left(\sum w_{ij} d(q, q_j)^2 + \lambda R(q) \right)$$

Prehension

Past occasions are inherited based on their causal relevance (w_{ij}).

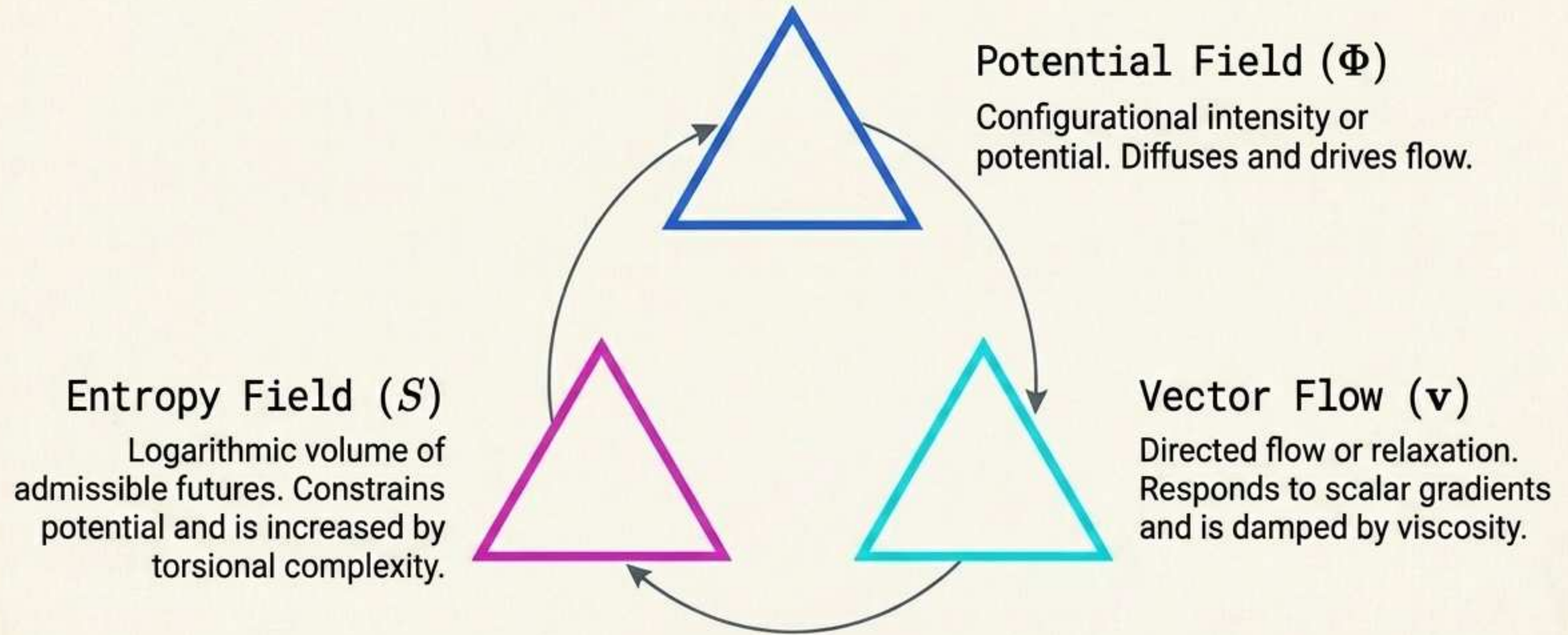
Concrescence

Reality is generated through local acts of integrating this inherited data.

Minimizing Tension

An actual occasion locally minimizes inherited causal tension, balanced by internal aesthetic coherence ($\lambda R(q)$).

Pillar 2: RSVP Dynamics—Translating Becoming into Continuous Fields

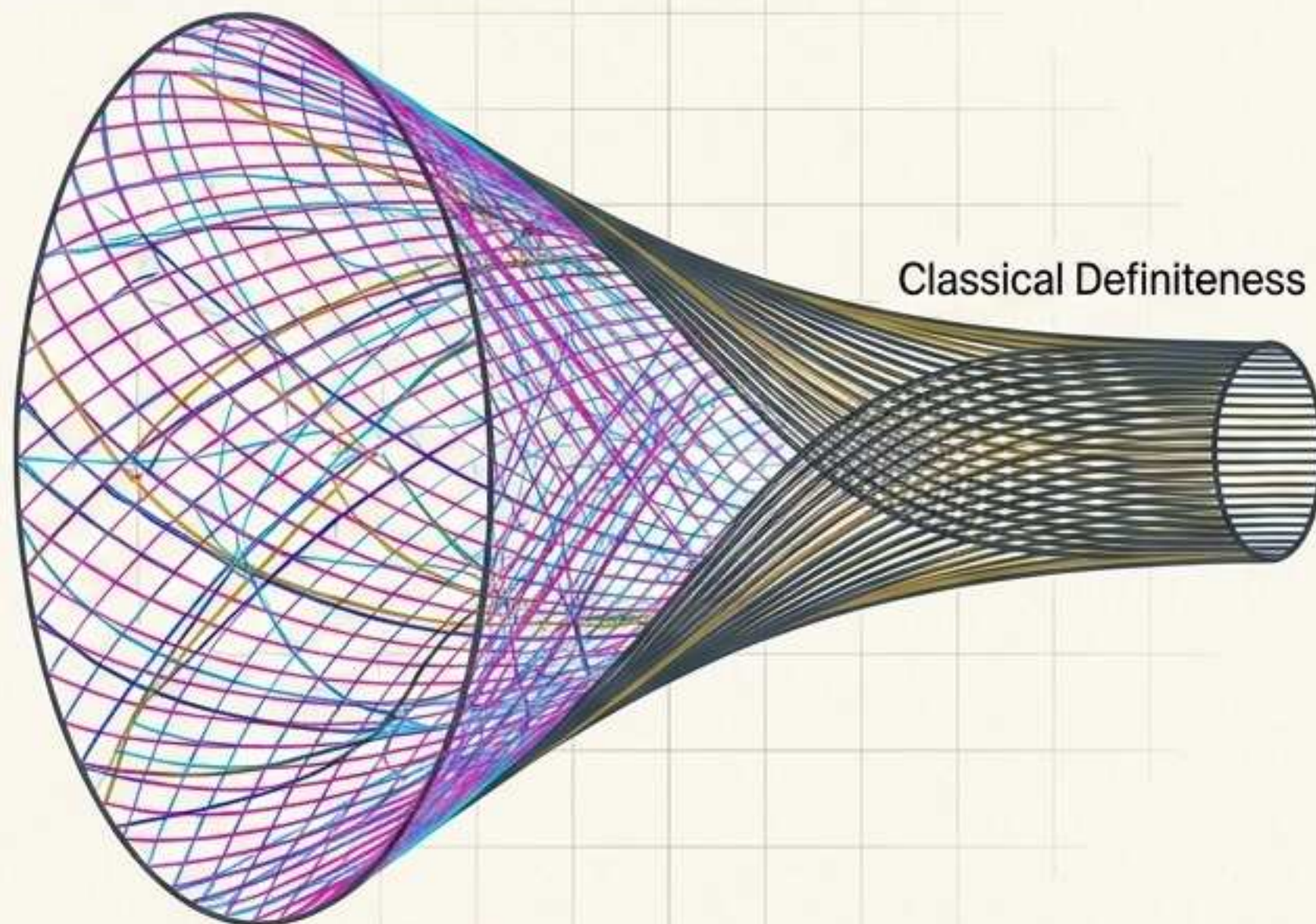


Insight: Whitehead's "actual occasion" appears in physics as a localized episode of this exact field relaxation—a temporary local attractor, not a permanent object.

The Geometry of Possible Futures: Effective Collapse Without Postulates

The Entropy-Accessibility Cone

Superposed Quantum Multiplicity



$$S(\mathbf{x}, t) = \log \text{Vol}(\Gamma(\mathbf{x}, t))$$

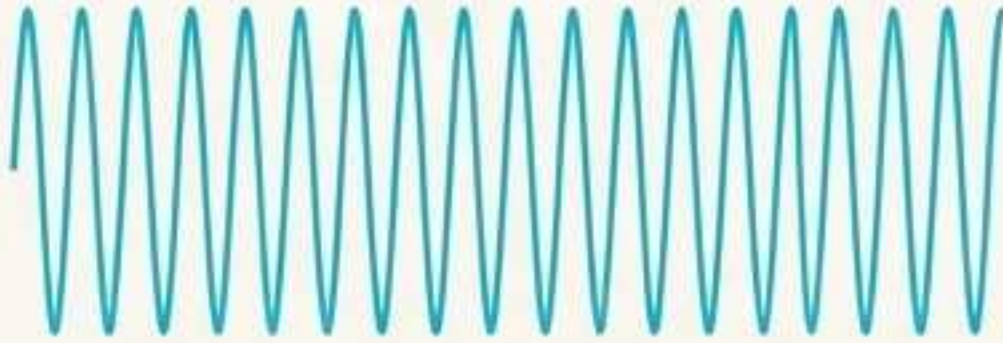
$$\partial_t \text{Vol}(\Gamma) < 0 \Rightarrow \partial_t S < 0$$

- **Accessibility Volume (Γ):**
The set of admissible future trajectories. High volume means many possible futures; low volume means constrained continuation.

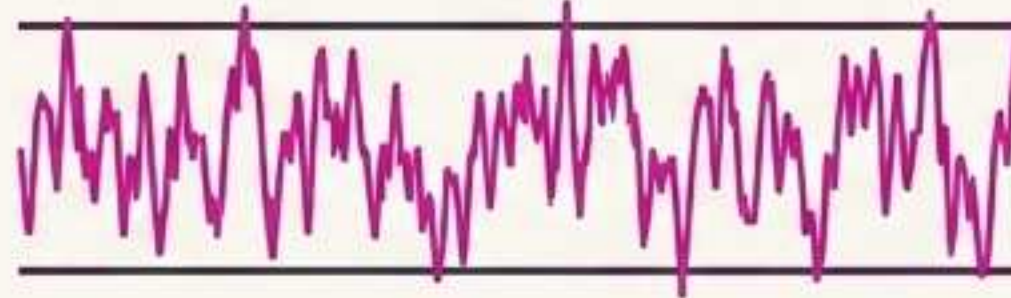
- **The Mechanism of Collapse:**
The physical world does not need to be metaphysically forced into one state. Instead, recursively recoverable alternatives simply vanish from the admissible future geometry.

Three Regimes of Accessibility and Signal Compression

Harmonic Lock



Bounded Stochastic Field



Impulsive Reset



Trajectory Volume: Low and contracting.

System State: Tightly constrained to a small basin. Deviation is near zero.

Signal Output: Absolute silence. Zero data stream required; highly predictable.

Trajectory Volume: Moderate and bounded.

System State: Chaotically distributed within a constraint envelope.

Signal Output: Bounded transmission. Output only needed to maintain envelope constraints.

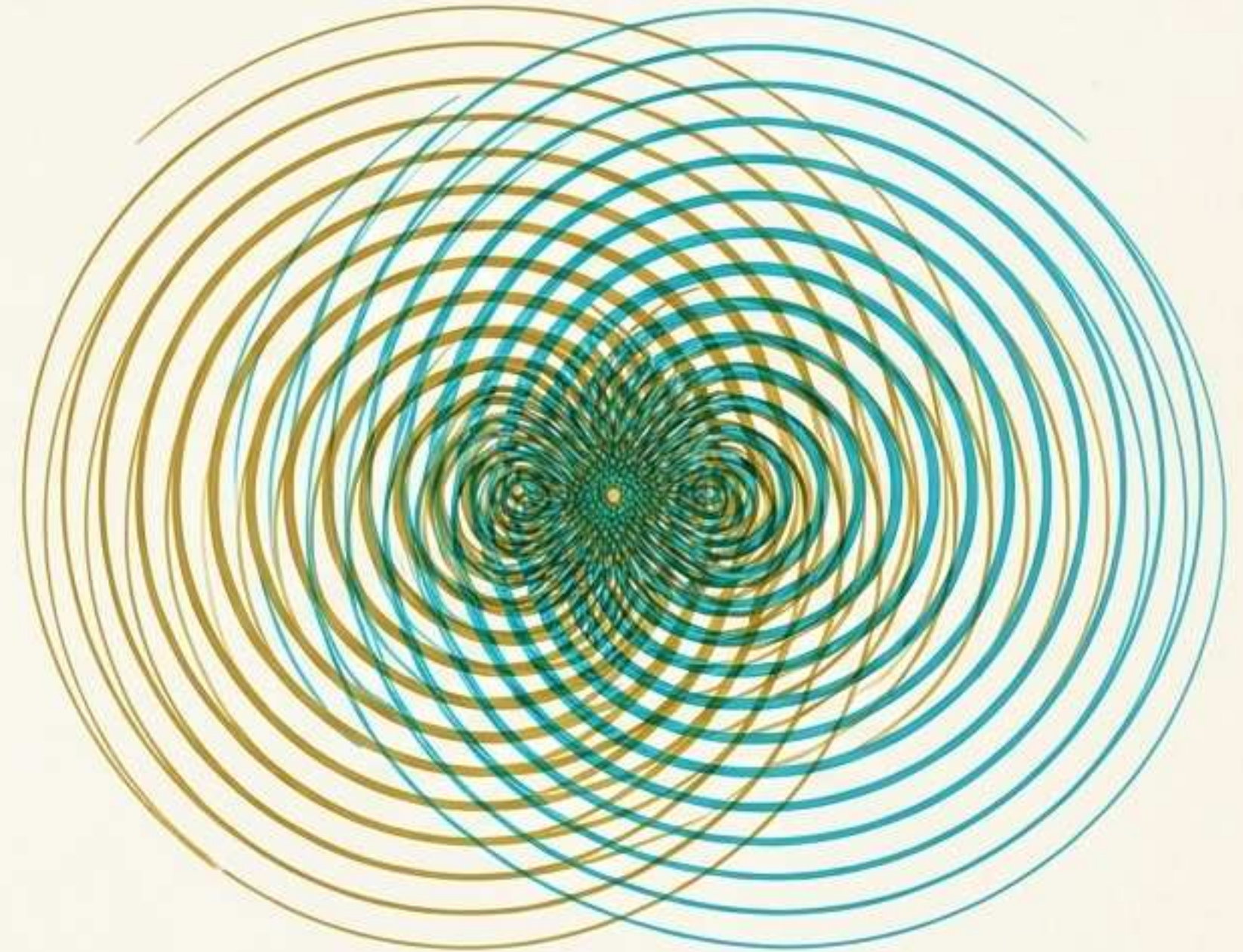
Trajectory Volume: Briefly maximizes, then collapses.

System State: Sudden transient event resetting the admissibility structure.

Signal Output: Brief, intense transmission.

The Wave Structure of Matter: Particles as Stabilized Interference

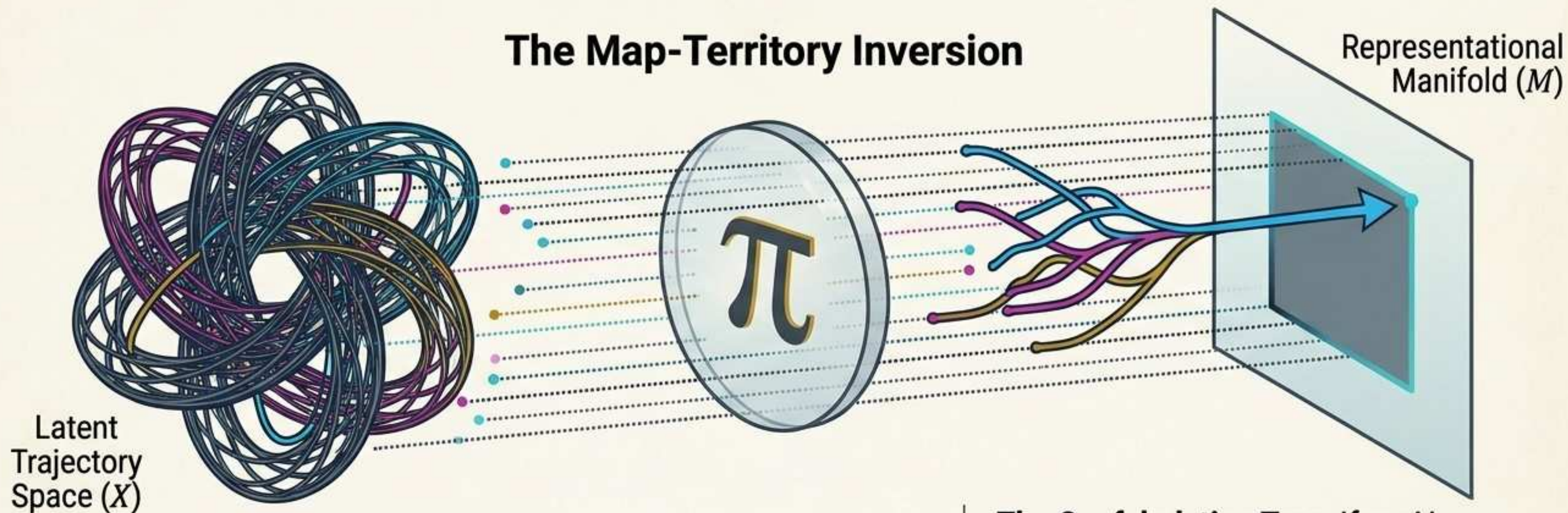
1. **The Particle Illusion:** Conventional physics treats matter as miniature billiard balls. WSM reveals particles as self-reinforcing interference structures—standing wave centers within a continuous medium.
2. **Periodic Recoverability:** Matter persists not because it is made of hard substance, but because its propagative field structure undergoes periodic recoverability ($\Psi(t + \tau) = e^{i\theta}\Psi(t)$).
3. **The RSVP Upgrade:** RSVP goes further. It insists the deeper object is not the wave itself, but the recursive accessibility structure by which wave-like phenomena emerge.



$$\Phi(\mathbf{r}, t) = \frac{2A_0}{r} e^{i\omega t} \cos(kr)$$

Pillar 3: CLIO & The Projection of Process

The Map-Territory Inversion

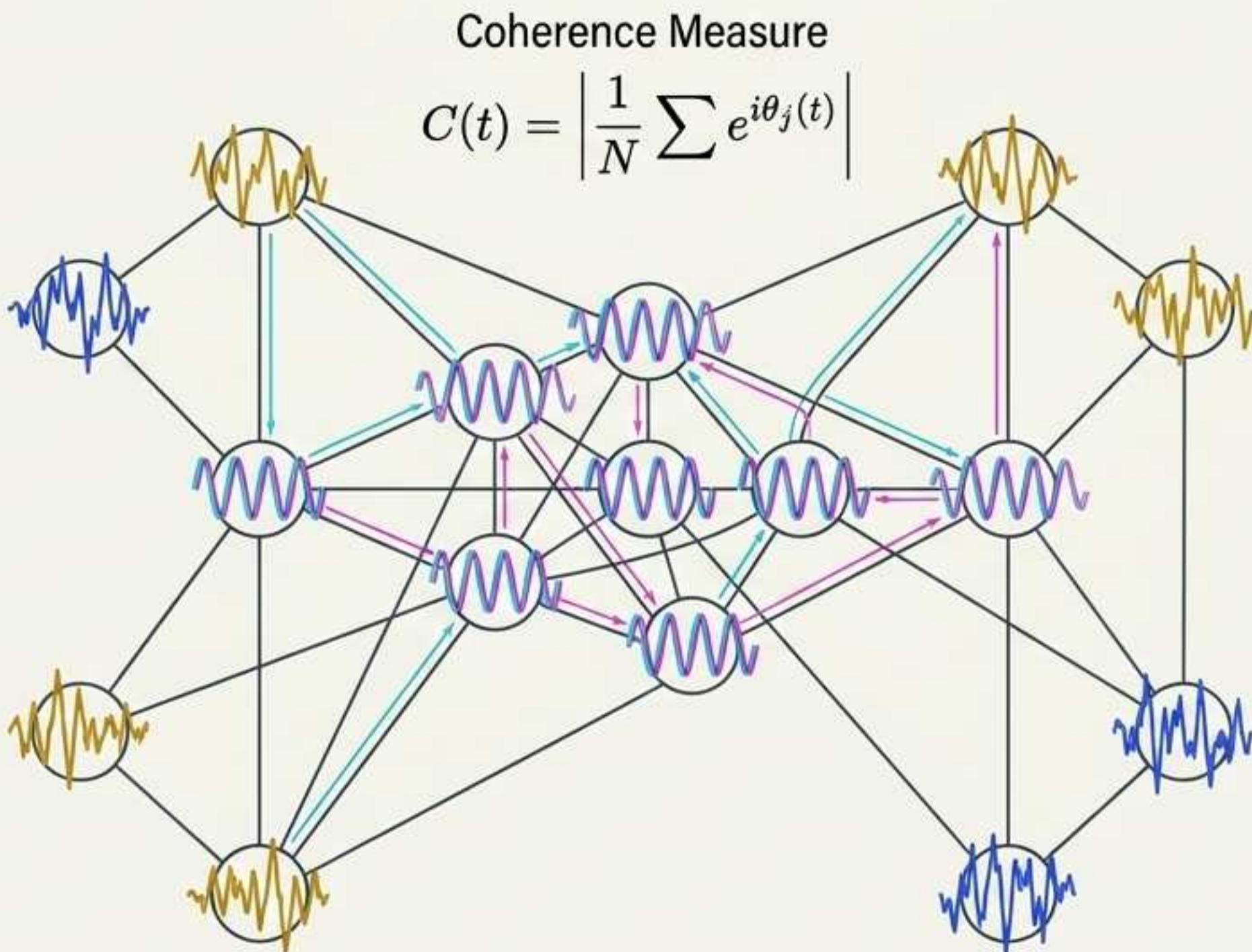


The Projection Map: $\pi: X \rightarrow M$ maps full, complex physical trajectories into compressed symbolic representations.

Representational Entropy: $S_\pi(m) = \log \text{Vol}(\pi^{-1}(m))$. High entropy means high ambiguity.

The Confabulation Trap: If an AI reasoning system optimizes only for the 2D shadow (the symbolic trace), it commits the Fallacy of Misplaced Concreteness. Multiple contradictory 3D processes map to the identical 2D narrative.

Pillar 4: MEM|8 and Propagative Memory Networks



Phase Alignment, Not Storage

Memory is not an inert symbol saved on a disk. It is a recursively regenerated nexus of propagative coherence. It persists when $C(t) \approx 1$.

Hierarchical Time

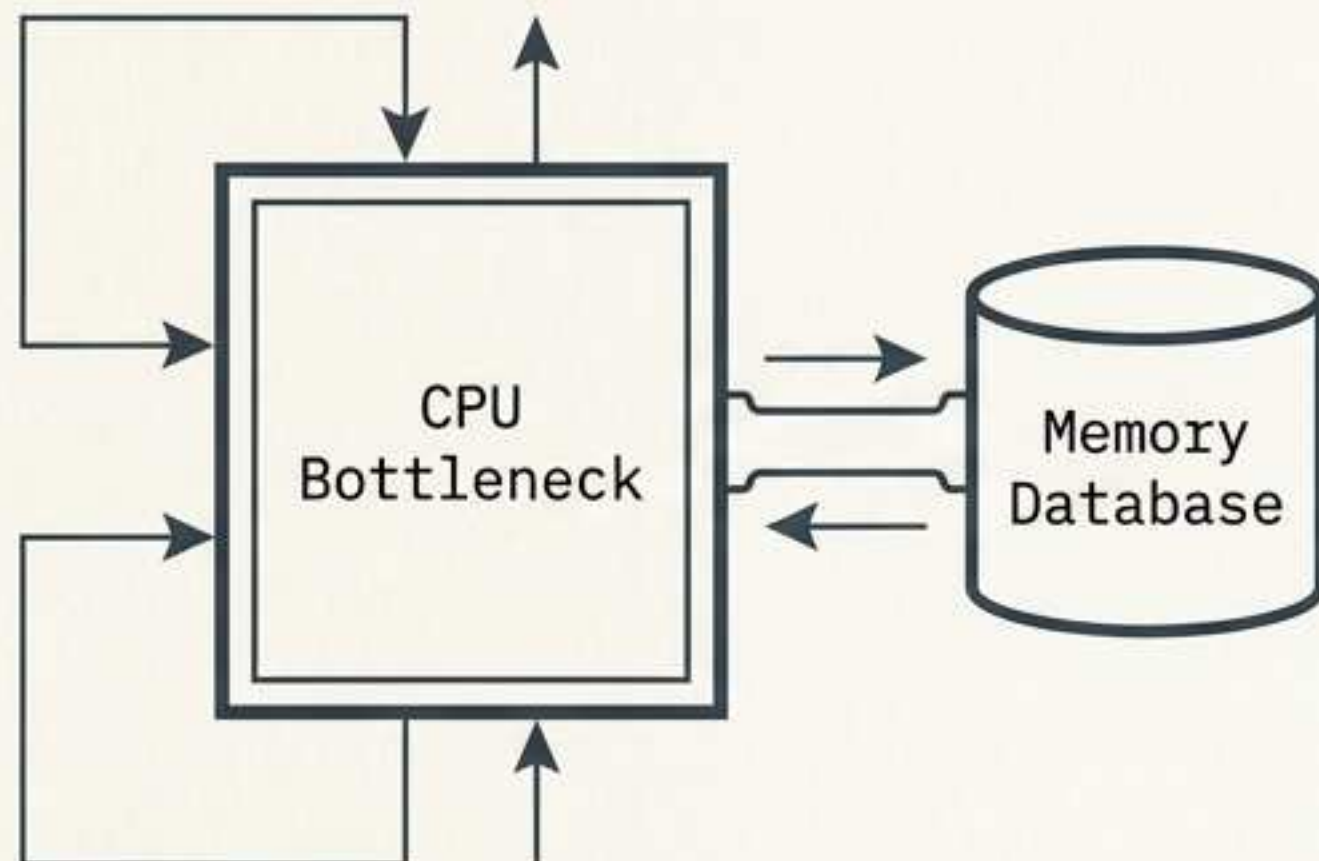
Nodes couple as parents and children. Temporal structure is encoded by relative phase offsets, not by a rigid global clock.

Salience

A memory trace is highly salient when it maintains energetic amplitude while minimizing incoherent phase noise.

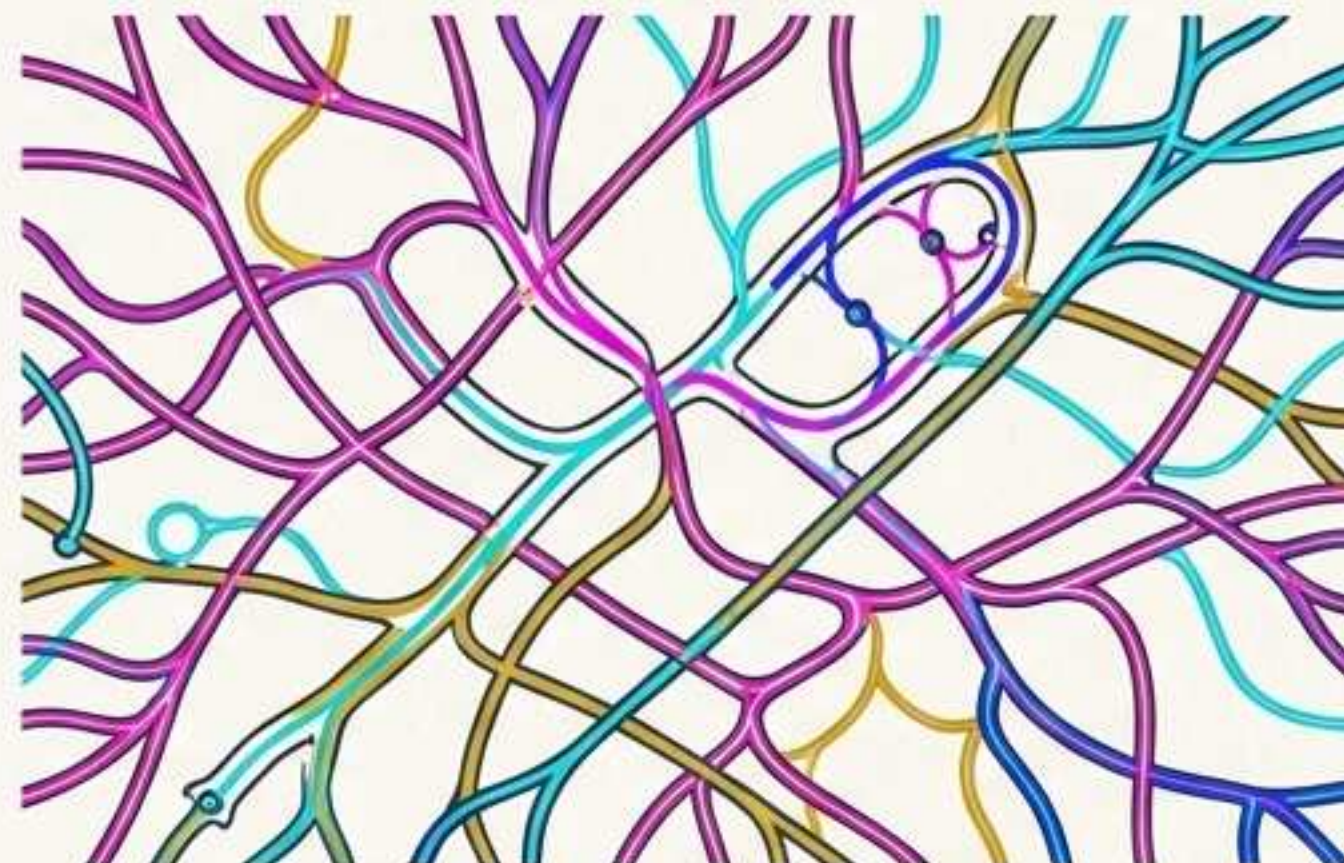
Rethinking Computation: Sequence vs. Propagation

The Sequential Paradigm (Turing/Von Neumann)



Ordered execution of instructions over centralized mutable state.

The Propagative Paradigm (MEM|8)



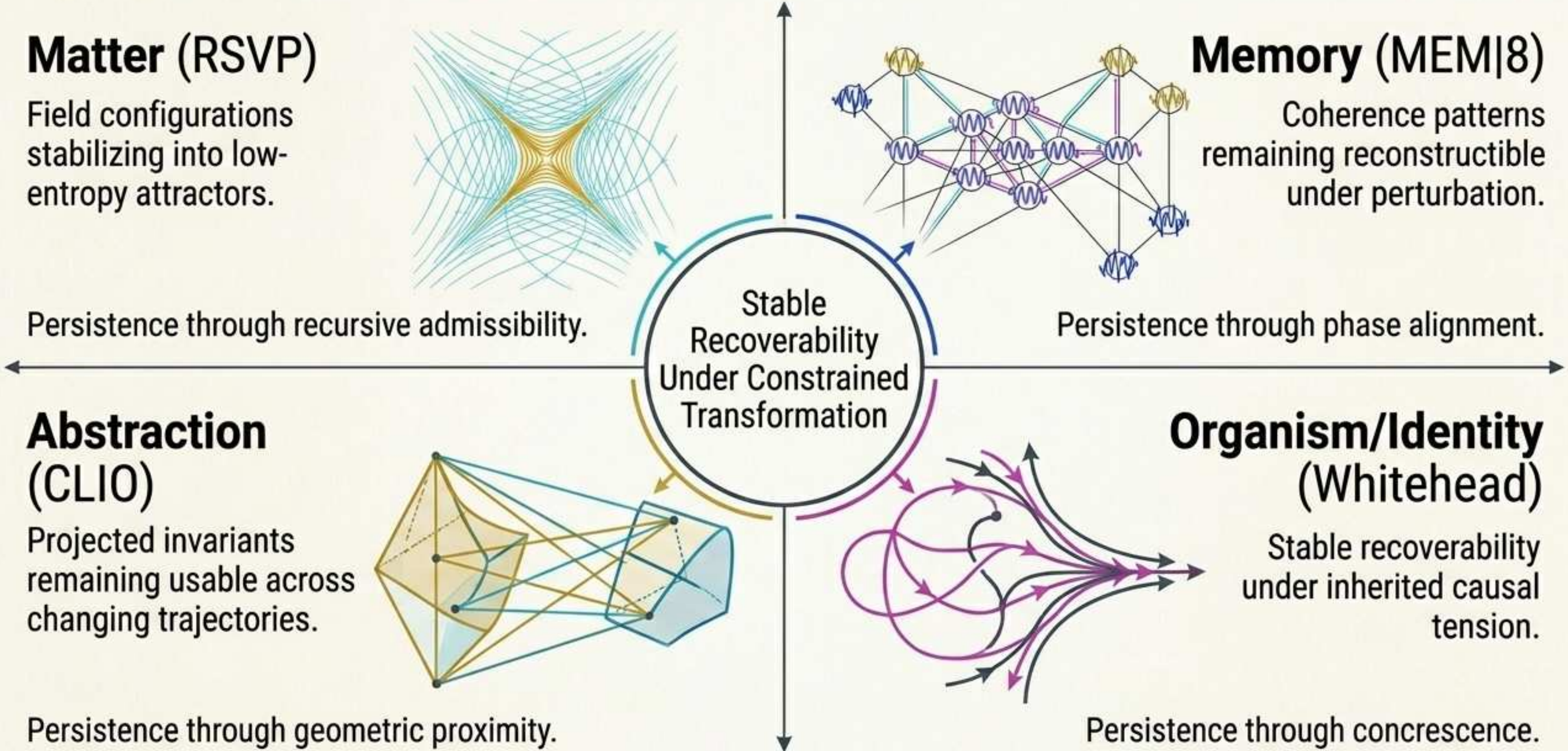
Constrained propagation across admissibility manifolds.

- Control Flow becomes Activation Flow.
- Memory Locality becomes Geometric Proximity.

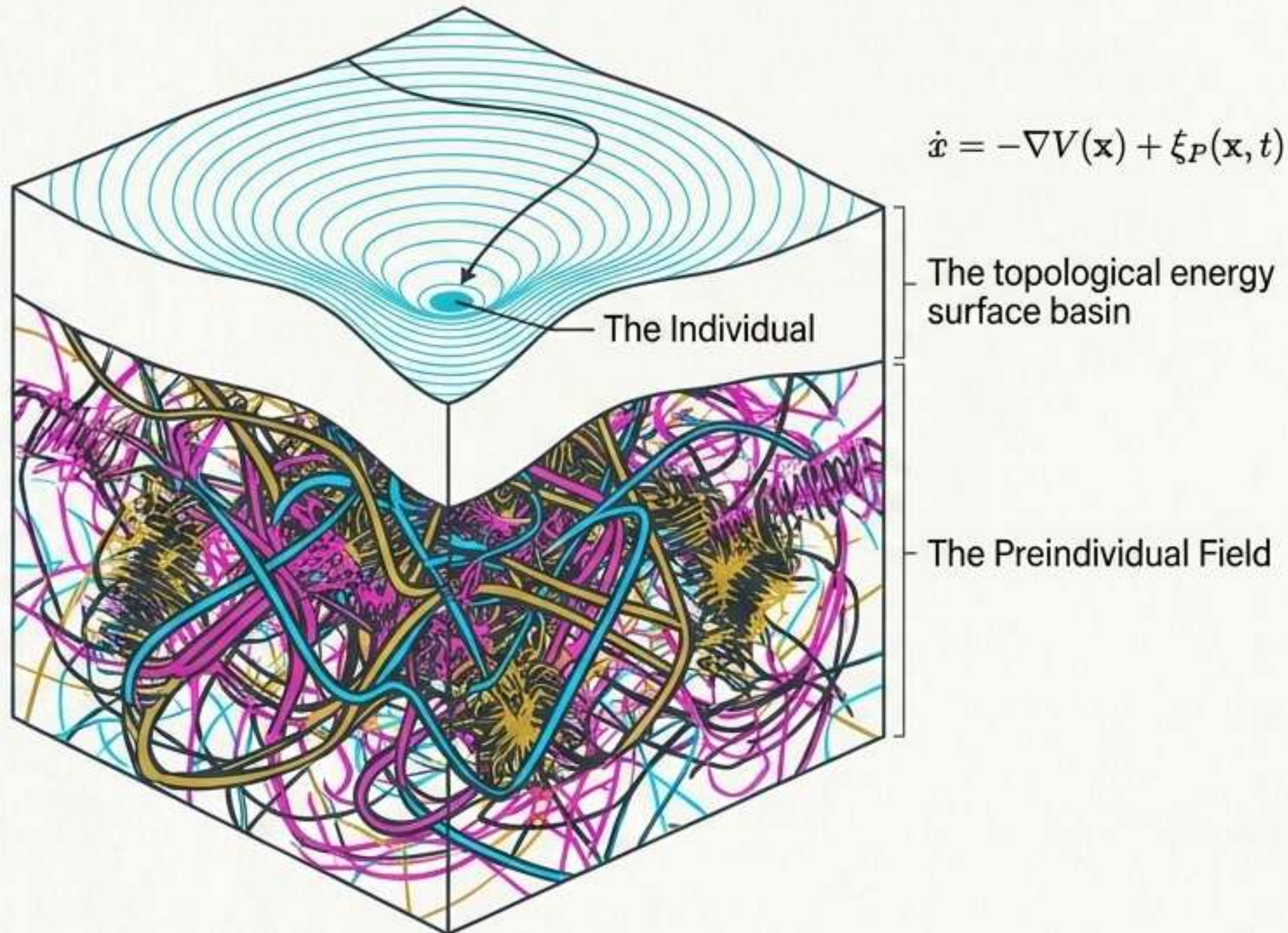
- Synchronization becomes Compatibility of Propagation Fronts.
- Scheduling becomes Manifold Navigation.

Computation is a structured activation geometry—thermodynamically sparse, concentrating energy only at active frontiers.

Proposition 1: The Periodic Table of Process



Pillar 5: Metastability and the Preindividual Field



The Simondonian Correction

Stabilization never exhausts the preindividual field. The individual is not a permanent object, but a temporary metastable resolution of deeper tensions.

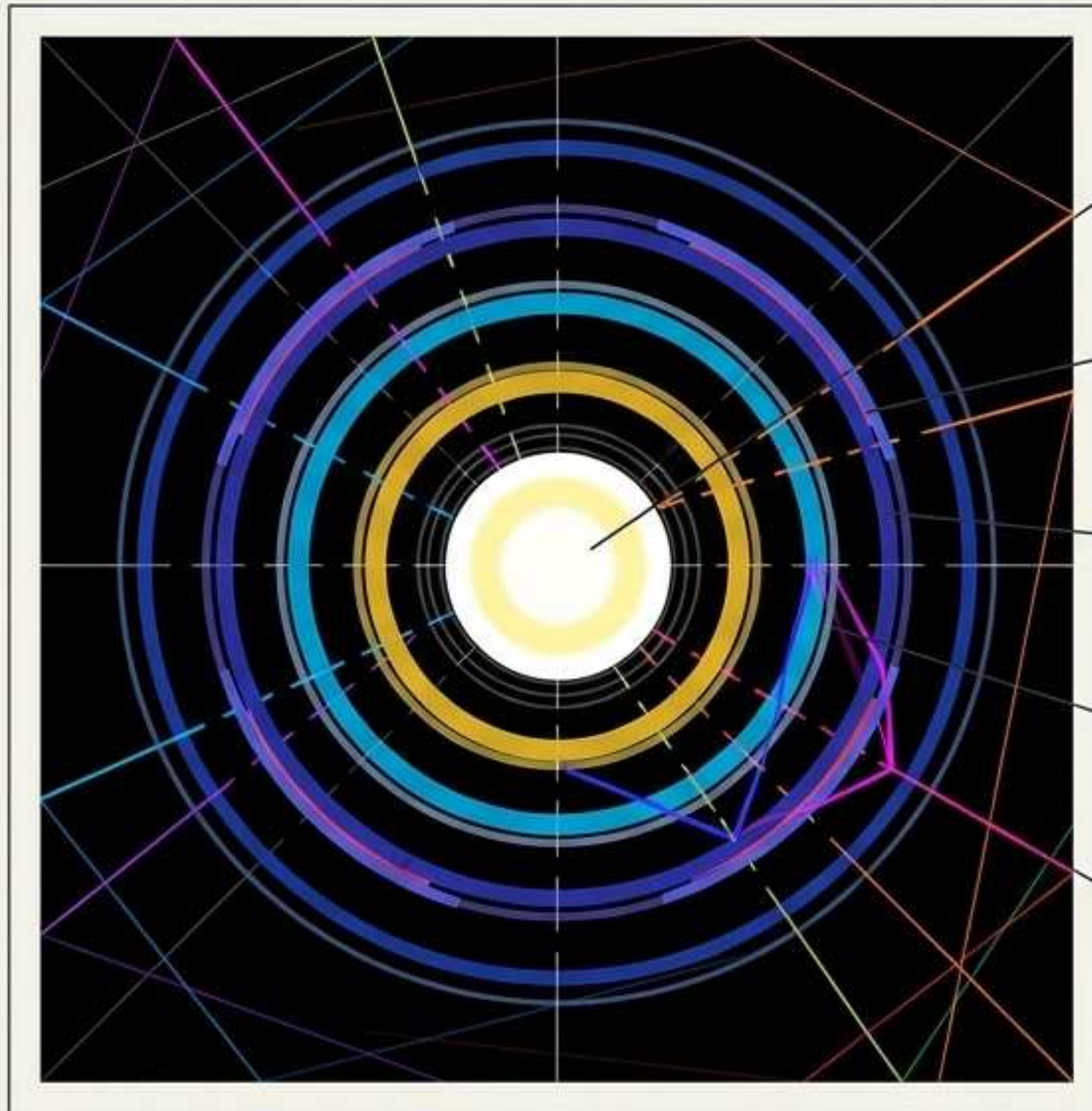
Organized Indeterminacy

The noise term ξ_P is not random thermal noise. It is latent organization, hidden symmetries, and unactualized relational tensions awaiting conditions to resolve into harmonic lock.

Thermodynamic Sparsity

Metastable individuals achieve coherent organization at extremely low thermodynamic cost by keeping the surrounding field quiescent.

Anatomy of a Metastable Field Individuation



Luminous Core: The central singularity. A high-coherence, low-entropy-accessibility zone.

Nested Stabilization Shells: Concentric rings matching the low-eigenvalue modes of the RSVP Laplacian.

Nodal Zones: The dark gaps between rings, representing destructive interference and near-zero accessibility.

Refractive Interference Rays: Propagative energy emitted from the core, reflecting off the boundary, and reinforcing the standing wave.

Anisotropic Constraint Boundary: The outer fourfold geometry that breaks continuous rotational symmetry into discrete channels.

The Unified Variational Principle

(RSVP)
Minimizing field tension.

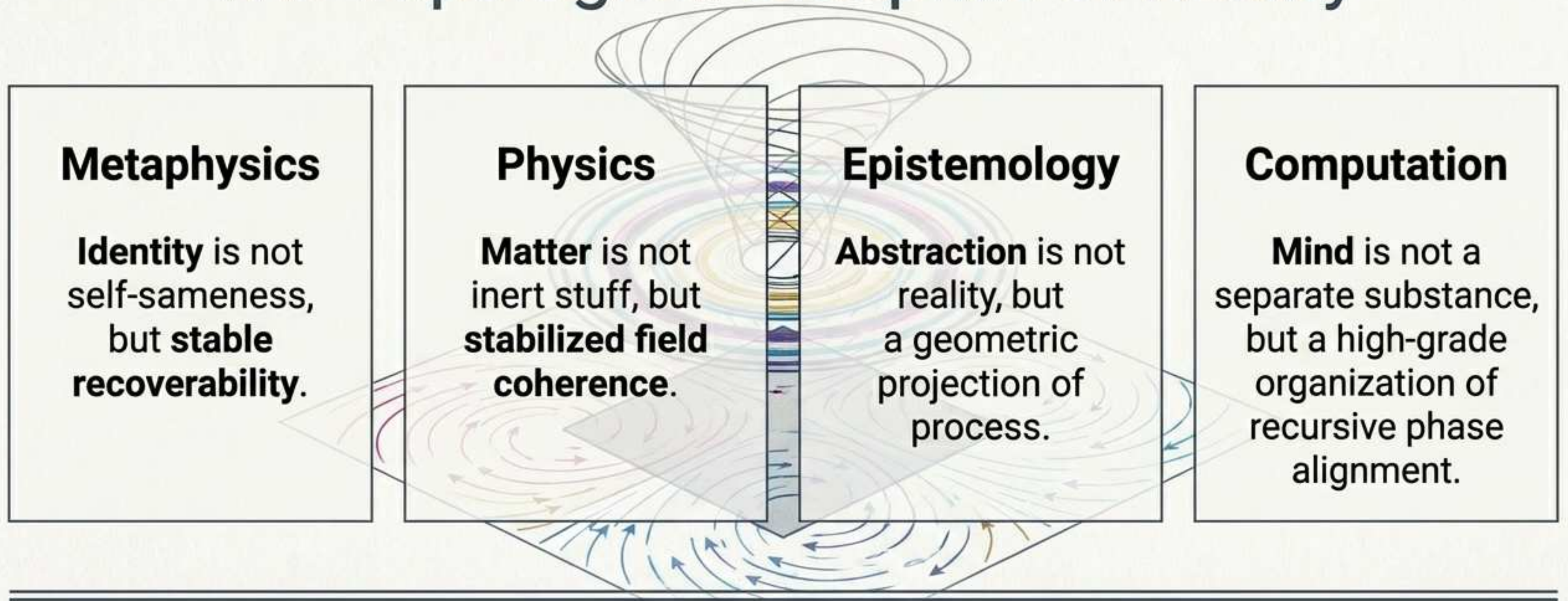
(CLIO)
Minimizing representational distortion.

$$S[h] = \int_0^T \left(E_{\text{constraint}} + E_{\text{projection}} + E_{\text{incoherence}} - \lambda N_{\partial}(h) \right) dt$$

(MEM|8)
Minimizing phase and
causal instability.

(Whitehead/Simondon) Constrained Novelty.
Pure tension minimization leads to static death. The
universe requires creative advance—rewarding
deviation that arises exactly at the metastable
boundaries of established coherence.

The Topological Blueprint of Reality



Reality is a metastable field of individuation in which stability and novelty continually co-produce one another.